Early-Pregnancy Blood Gene Expression, Maternal Vitamin D Status, and Preeclampsia

Jason Spino

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1 Methods

1.1 Data source

Data come from a nested case-control study within the Vitamin D Antenatal Asthma Reduction Trial (VDAART), described by Mirzakhani et al. [1] and deposited in the Gene Expression Omnibus as series GSE85307 [2]. Peripheral blood was collected at 10 to 18 weeks of gestation, and RNA was profiled on the Affymetrix Human Gene 1.0 ST array. The initial dataset contained 157 women, 47 of whom later developed preeclampsia and 110 of whom had a normal pregnancy. Three files were used: a covariate file, an expression matrix with one row per array probe and one column per sample, and a linker file matching probe IDs to gene names. Samples were matched between files using the GEO sample ID (geoID), and the match was checked to confirm that the expression and covariate files contained exactly the same 157 samples.

1.2 Final sample size

No covariate had missing values and every probe matched a gene name, so no samples were excluded. The final sample size was 157 women (110 normal, 47 preeclampsia).

1.3 Variables

Three genes were studied in detail: *RUNDC3A*, *LYN*, and *IGFBP1*. The continuous covariate was maternal body mass index (BMI, kg/m^2). The two categorical covariates were pregnancy outcome (normal or preeclampsia) and maternal vitamin D status, which was derived from baseline 25-hydroxyvitamin D. Women with 25-hydroxyvitamin D below 20 ng/mL were classified as deficient and the rest as sufficient. Two further continuous covariates, maternal age and gestational age at delivery, were included in the summary statistics table.

1.4 Analyses

Participant characteristics were summarised by pregnancy outcome (Table 1). Continuous covariates are shown as mean (SD) and were compared with Welch two-sample t-tests. Vitamin D status is shown as n (%) and was compared with a chi-squared test with continuity correction.

For each gene, a histogram showed the distribution of expression, a scatterplot with a least-squares line showed expression against maternal BMI (summarised by the Pearson correlation

coefficient r), and a boxplot showed expression by pregnancy outcome and vitamin D status together. These nine plots were produced by a single R function that takes a data frame and a vector of gene names and loops over the genes.

For the heatmap, the 10 genes with the largest standard deviation across all 157 samples were selected. This choice did not use pregnancy outcome or vitamin D status. Expression was converted to z-scores within each gene, and both genes and samples were clustered hierarchically using Euclidean distance and complete linkage. Tracking bars show pregnancy outcome and vitamin D status. To check whether the sample clusters related to outcome, the sample dendrogram was cut into two clusters and compared with pregnancy outcome using a chi-squared test.

As an additional plot, a forest plot shows adjusted associations for the three genes. For each gene, a linear model was fit with standardised expression as the outcome and pregnancy outcome, vitamin D status, and standardised maternal BMI as predictors. Because expression and BMI were both standardised, every estimate is in standard deviations of gene expression, which puts all three genes on the same scale. Estimates are shown with 95% confidence intervals. Significance was set at $p < 0.05$, with no correction for multiple comparisons.

1.5 Software

All analyses were done in R version 4.6.1 [3]. Data were cleaned and reshaped with the tidyverse [4] (version 2.0.0), including dplyr 1.2.1 [5], tidyr 1.3.2 [6], readr 2.2.0 [7], stringr 1.6.0 [8], purrr 1.2.2 [9], and tibble 3.3.1 [10]. Plots were made with ggplot2 4.0.3 [11, 12] and the heatmap with pheatmap 1.0.13 [13]. Table 1 was generated with knitr 1.52 [14] and kableExtra 1.4.1 [15], and the analysis was written in R Markdown using rmarkdown 2.32 [16].

1.6 Code availability

All code is available at `https://github.com/JasonOSpino/HSE103-Project`.

2 Results

2.1 Participant characteristics

Women who developed preeclampsia had a higher mean BMI than women with a normal pregnancy (31.2 vs. 28.0 kg/m^2, $p = 0.02$) and delivered earlier (37.8 vs. 39.3 weeks, $p = 0.01$). Maternal age did not differ between groups ($p = 0.33$). Vitamin D deficiency was somewhat more common in the preeclampsia group (55.3% vs. 47.3%), but the difference was not significant ($p = 0.45$) (Table 1).

Table 1: Participant characteristics by pregnancy outcome. Continuous variables are mean (SD) and compared with Welch t-tests. Vitamin D status is n (%) within each outcome group and compared with a chi-squared test.

Characteristic	Overall (n = 157)	Normal (n = 110)	Preeclampsia (n = 47)	p-value
Maternal BMI (kg/m^2)	28.9 (7.9)	28.0 (7.7)	31.2 (8.0)	0.02
Maternal age (years)	26.4 (5.0)	26.6 (5.1)	25.8 (4.9)	0.33
Gestational age at delivery (weeks)	38.8 (2.2)	39.3 (1.0)	37.8 (3.5)	0.01
Vitamin D status, n (%)				0.45
Deficient (<20 ng/mL)	78 (49.7)	52 (47.3)	26 (55.3)	
Sufficient (≥20 ng/mL)	79 (50.3)	58 (52.7)	21 (44.7)	

2.2 Distribution of gene expression

Expression of all three genes was roughly symmetric and unimodal. *RUNDC3A* had the widest spread (mean 9.04, SD 0.68) (Figure 1a), while *LYN* (mean 12.18, SD 0.24) (Figure 2a) and *IGFBP1* (mean 8.33, SD 0.15) (Figure 3a) varied much less between women.

2.3 Gene expression and maternal BMI

LYN expression decreased with higher maternal BMI ($r = -0.28$, $p < 0.001$) (Figure 2b). *RUNDC3A* was not associated with BMI ($r = 0.01$, $p = 0.89$) (Figure 1b), and neither was *IGFBP1* ($r = -0.10$, $p = 0.19$) (Figure 3b).

2.4 Gene expression by pregnancy outcome and vitamin D status

RUNDC3A expression was lower in women who developed preeclampsia, in both vitamin D groups (Figure 1c). *LYN* expression was higher in vitamin D sufficient women regardless of pregnancy outcome (Figure 2c). *IGFBP1* showed no clear difference across the four groups (Figure 3c).

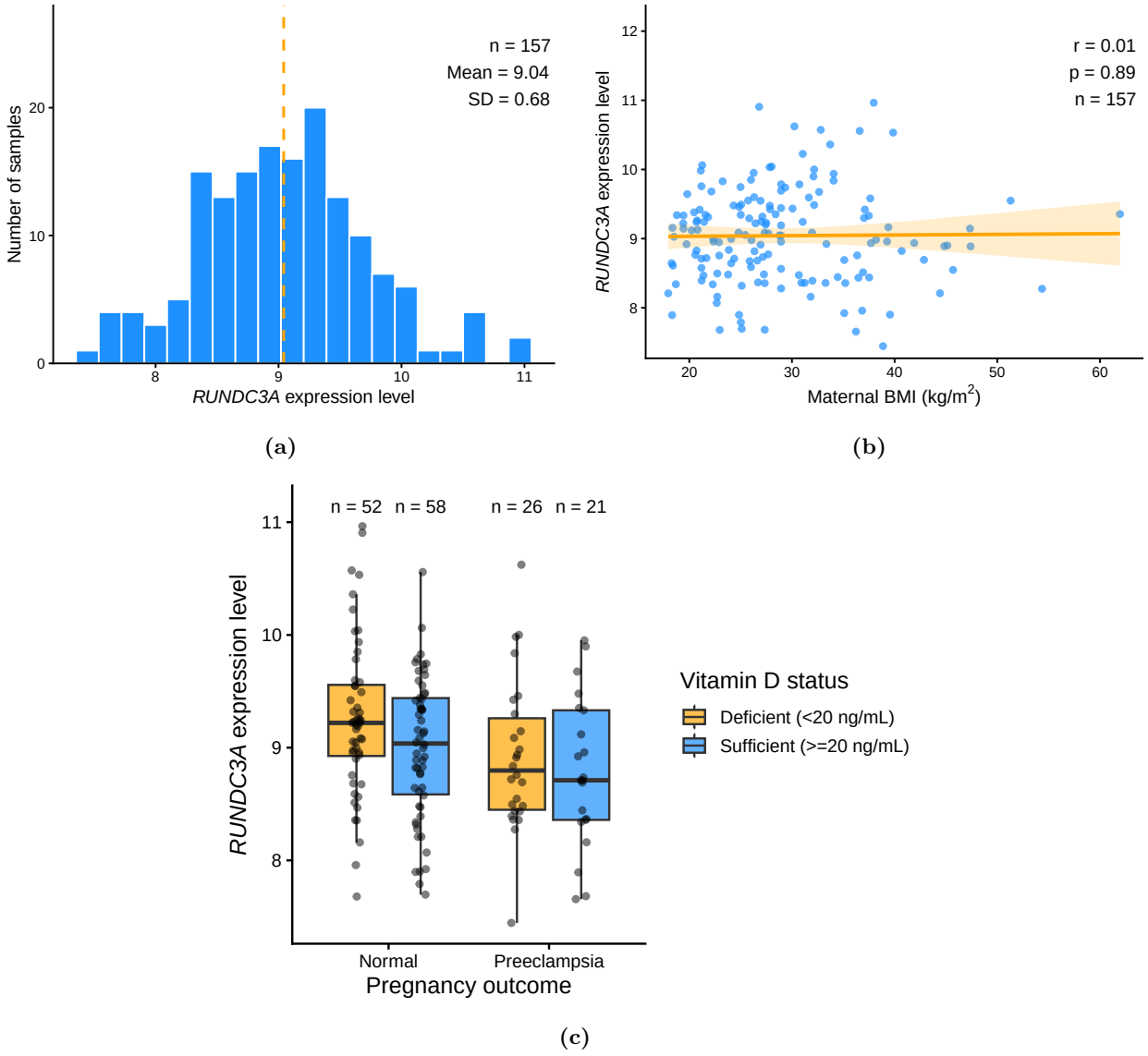


Figure 1: *RUNDC3A* expression ($n = 157$). (a) Distribution of expression; the dashed line marks the mean. (b) Expression against maternal BMI with a least-squares line and 95% confidence band; r is the Pearson correlation. (c) Expression by pregnancy outcome and vitamin D status; points are individual women and labels give group sizes.

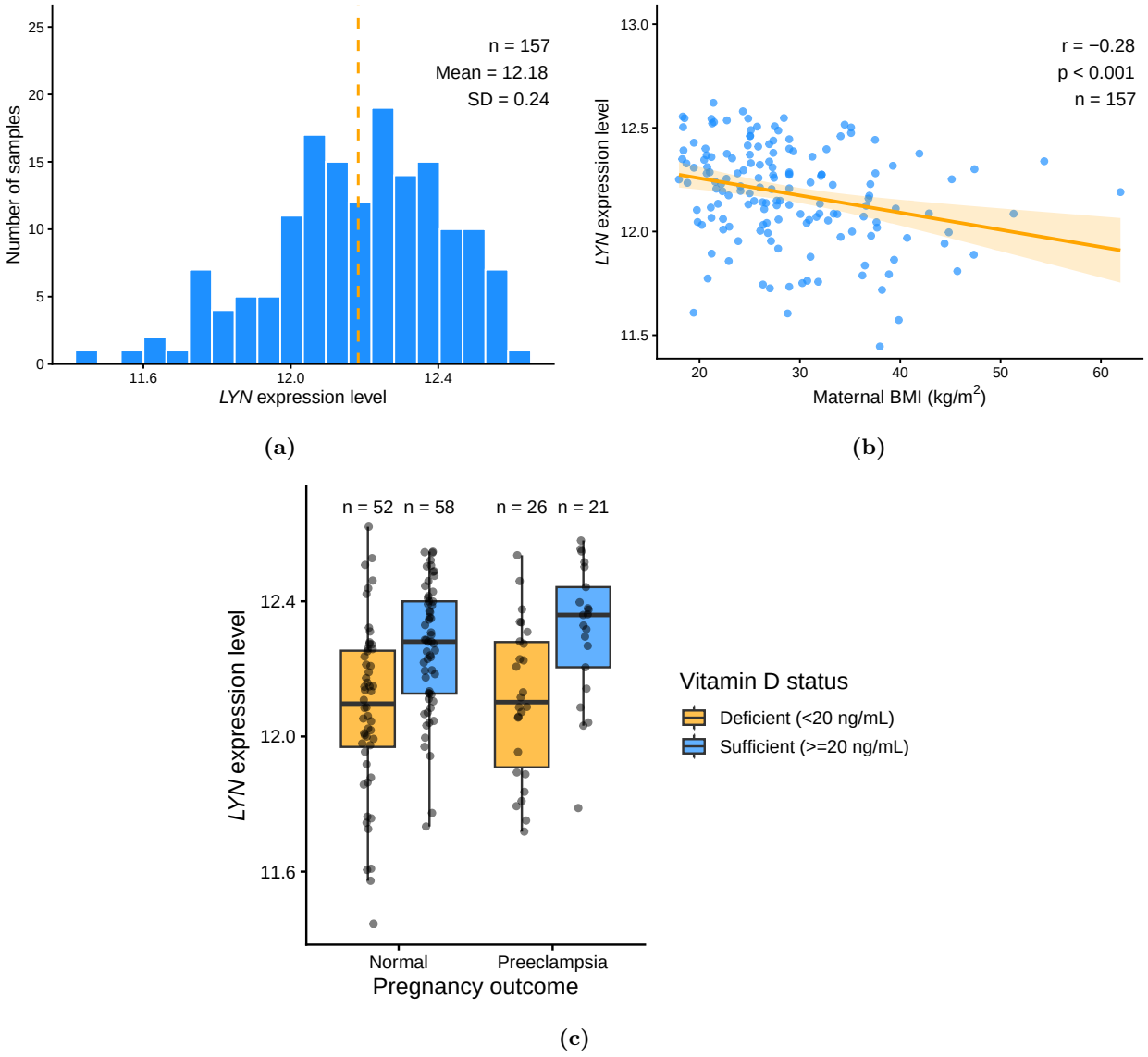


Figure 2: *LYN* expression ($n = 157$). Panels as in Figure 1: (a) distribution, (b) relationship with maternal BMI, and (c) expression by pregnancy outcome and vitamin D status.

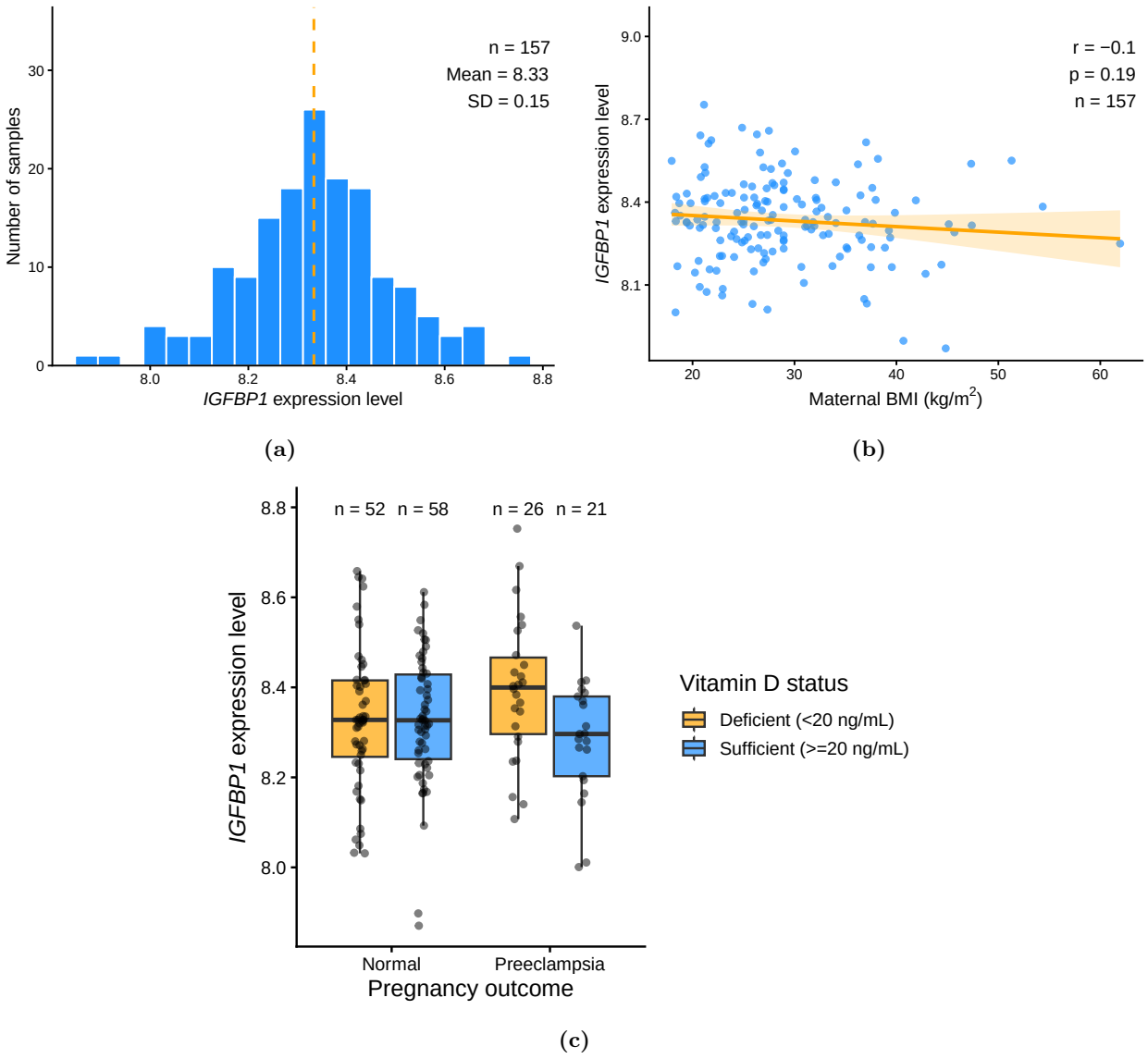


Figure 3: *IGFBP1* expression ($n = 157$). Panels as in Figure 1: (a) distribution, (b) relationship with maternal BMI, and (c) expression by pregnancy outcome and vitamin D status.

2.5 Heatmap of the 10 most variable genes

The 10 most variable genes were *UTS2*, *RUNDC3A*, *TMEM176B*, *CEACAM6*, *CLIC2*, *SLC1A5*, *ZNF594*, *ZBTB41*, *CAPZA2*, and *TGM2*. Clustering on these genes did not separate women by pregnancy outcome. When the sample dendrogram was cut into two clusters (136 and 21 women), the proportion who developed preeclampsia was similar in each (30.9% vs. 23.8%, $p = 0.69$) (Figure 4).

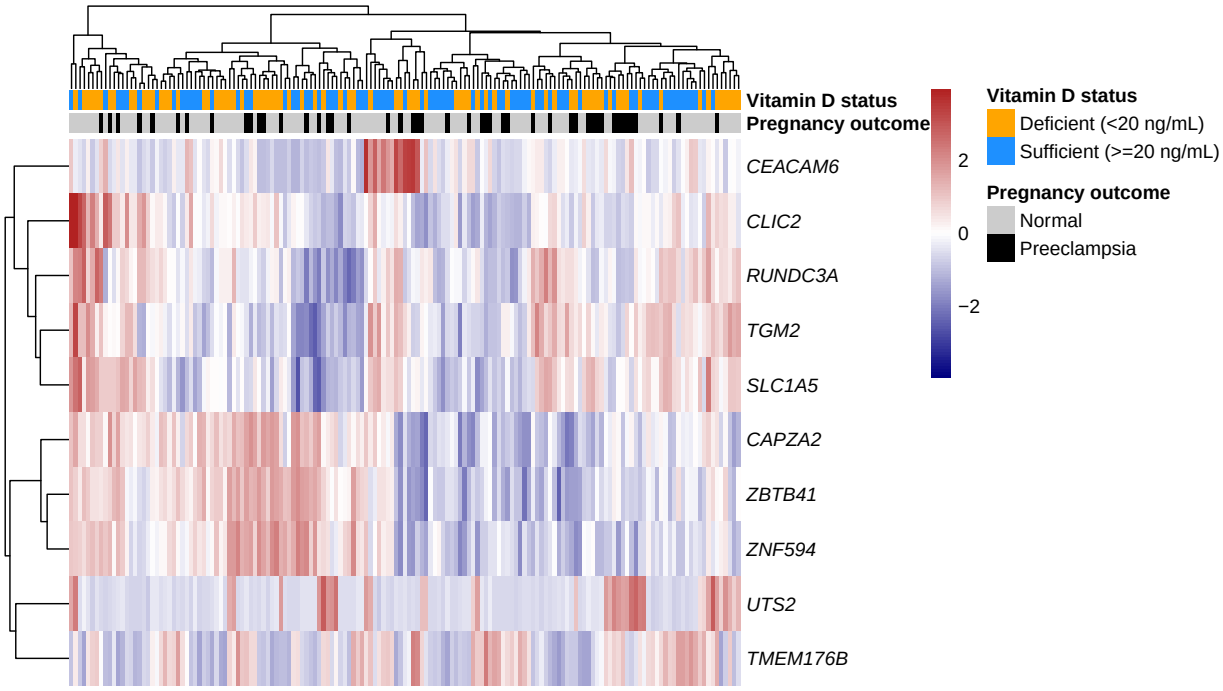


Figure 4: Hierarchically clustered heatmap of the 10 genes with the largest standard deviation across all 157 women. Values are z-scores within each gene (red above that gene's mean, blue below). Genes (rows) and women (columns) are clustered by Euclidean distance with complete linkage. Tracking bars show pregnancy outcome and vitamin D status.

2.6 Adjusted associations

After adjusting for each other, preeclampsia was associated with lower *RUNDC3A* expression (-0.41 SD, 95% CI -0.75 to -0.07, $p = 0.02$), and vitamin D sufficiency was also associated with lower *RUNDC3A* (-0.33 SD, 95% CI -0.65 to -0.01, $p = 0.045$). *LYN* showed the largest effect in the model: vitamin D sufficient women had expression 0.68 SD higher (95% CI 0.38 to 0.98, $p < 0.001$), and each 1 SD increase in BMI was associated with 0.20 SD lower expression (95% CI -0.35 to -0.04, $p = 0.01$). No covariate was significantly associated with *IGFBP1*, although vitamin D sufficiency ($p = 0.06$) and BMI ($p = 0.06$) were close to the threshold (Figure 5).

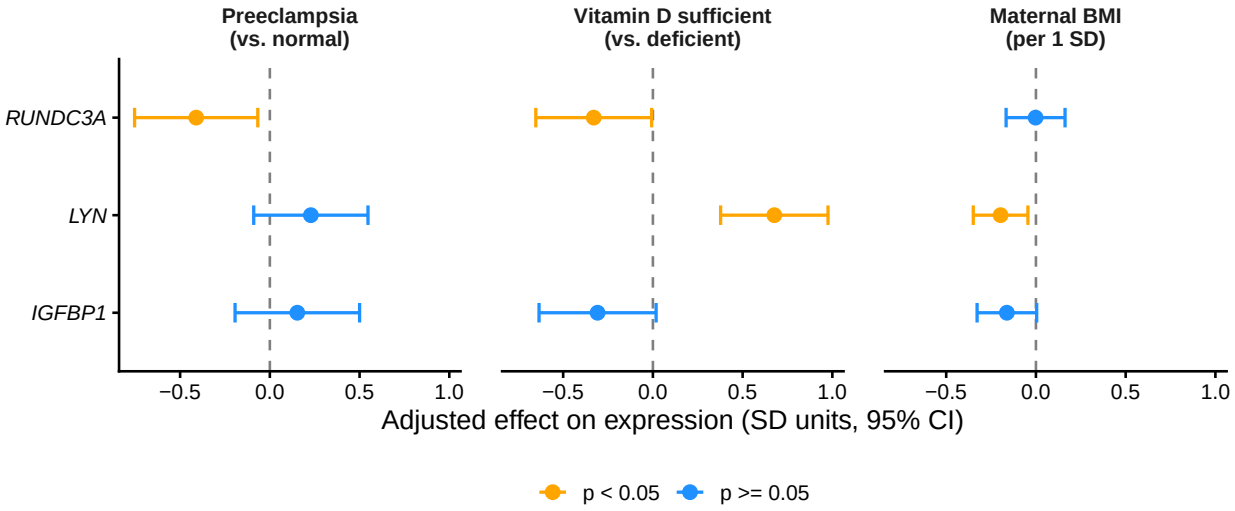


Figure 5: Forest plot of adjusted associations with gene expression. For each gene, one linear model included pregnancy outcome, vitamin D status, and maternal BMI. Expression and BMI were standardised, so estimates are in standard deviations of expression. Points are estimates and bars are 95% confidence intervals; orange marks $p < 0.05$. The dashed line marks no effect.

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